

Dexter Fernandes

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https://github.com/Dexter-Fernandes

Software Engineer

Computer Vision Engineer with 3+ years of experience delivering low-latency vision solutions for Tier 1 companies, including a construction firm in Texas and a UK oil and gas company. Proven track record of collaborating with multinational clients to deploy AI-driven vision solutions in fast-paced dynamic environments. Experienced in applying state-of-the-art AI research while leading testing, QA, and monitoring processes.

PROJECTS

FinanceBench RAG Evaluation Harness

Jun 2026

- Built an evaluation framework for FinanceBench, a financial question-answering benchmark, to measure how well RAG and document-grounded QA systems answer questions using the right source evidence.
- Designed structured experiments across closed-book, oracle-context, and dense RAG workflows to identify whether failures came from retrieval, source selection, or answer generation.
- Added LLM-as-judge scoring, evidence matching, citation checks, numeric matching, and hallucination labels to make AI agent performance easier to inspect, compare, and improve.
- Produced reproducible runs using automated tests, config-driven experiments, structured prediction outputs, and Markdown reports, enabling regression checks and clearer technical recommendations for improving retrieval and answer quality.

DeepStream RTSP Pipeline

Jun 2026

- Built an end-to-end **NVIDIA DeepStream 9.0/GStreamer RTSP video analytics pipeline** for multi-stream computer vision, using rtspsrc, nvstreammux, nvinfer, nvtracker, and nvdsosd to run YOLO+NvDCF tracking, metadata extraction, anonymisation, CSV logging, and re-streamed output across 3 concurrent camera feeds.
- Deployed and optimised a **YOLO26n TensorRT FP16 detector** on **6GB VRAM Turing hardware**, combining dynamic-batch engine conversion, nvinfer integration, p99 latency profiling, and FP16-vs-FP32 accuracy validation to sustain **29.7 FPS per stream live** and **131.3 FPS per stream unthrottled**.
- Implemented a **C++/CUDA TensorRT decode plugin** using IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Benchmarked tracker configurations across **IOU, NvDCF, and** ByteTrack-inspired **NvSORT**, producing MOTA, IDF1, ID switches, fragmentation, per-frame detection counts, FPS, and VRAM impact to guide production tracker selection.
- Added production hardening through **GPU smoke tests, model-promotion gates, structured JSON logging, and per-camera health metrics** for reactive debugging across liveness, dropped frames, reconnect events, and RSS.

WORK EXPERIENCE

GrowthStage • Remote

May 2026 - Present

Recruiter

- Evaluated candidates' technical depth across perception, SLAM, and deployment using hands-on robotics and computer vision knowledge to separate genuine signal from surface-level claims.

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.

- Orchestrated a sensor-data cleansing pipeline for 3.6 million records using Pandas and Apache Airflow, reducing model rollout time from 3 days to under 24 hours.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts reducing live video pipeline crashes by 67% and improved root-cause analysis.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, improving maintainability and reducing defects.
- Documented deployment guardrails, failure modes, and recovery procedures, improving maintainability for engineers supporting fielded systems.
- Implemented continuous integration and validation workflows for edge inference modules for 24/7 deployments to improve release consistency across hardware targets.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.
- Integrated AI modules with production systems in collaboration with other teams to deliver fault-tolerant edge deployments.
- Engineered production-grade AI software for real-time perception and safety analytics, delivering deterministic performance across 24/7 oil rig operations.
- Validated deterministic model performance under extreme field conditions at a British Petroleum site, ensuring reliability across 24/7 operations.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Led on-site validation and diagnostics on deployments, troubleshooting model failures under environmental stressors and translating performance findings into fixes that improved robustness in live operation.
- Built synthetic image generation and augmentation procedures with OpenCV and scikit-image to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.
- Diagnosed and resolved production failures through on-site analysis and structured validation checklists, reducing crash frequency by 67% in live video pipelines.
- Built a Django-based internal computer vision application with a web dashboard to visualise inference, system health, and performance metrics, enabling faster debugging.
- Designed adversarial training experiments (occlusion, lighting variation, motion artefacts), benchmarking robustness via F1 and failure-rate metrics, achieving a 15% F1-score improvement within 3 months.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.
- Built benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, EfficientDet) and quantify accuracy, latency, and stability pre- and post-optimisation for regression detection.
- Diagnosed and resolved production issues in Dockerised microservices, improving service continuity through stronger logging, health checks, safe restarts, and workload isolation.
- Implemented reliability patterns and operational observability, using timeouts, retries, message validation, and structured logs and metrics to improve root-cause analysis and maintain stable behaviour under load.
- Added operational observability using structured logs and metrics, enabling faster root-cause analysis and documented failure modes and recovery procedures to reduce operational overhead.
- Developed an internal web dashboard in Django to surface inference outputs, system health, and performance metrics, enabling faster debugging and root-cause analysis across deployed systems.
- Led validation, benchmarking, and observability for fielded AI deployments, reducing crash frequency by 67% and documenting failure modes, recovery procedures, and deployment guardrails for engineers operating live systems.

- Applied image preprocessing and artefact-detection workflows with OpenCV and scikit-image to identify mud and sand contamination on camera feeds, reducing false positives by 13% in harsh industrial environments.
- Benchmarked and optimised edge inference workflows across latency, memory, and stability constraints, defining repeatable FP16 validation paths and acceptance thresholds for reliable deployment across hardware targets.
- Led field validation on BP oil-rig deployments, diagnosing model failures under lighting, motion and environmental stressors and translating findings into practical fixes for live operation.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Worked with multinational stakeholders, from project managers to drill rig operators, translating operational problems into deployable solutions and communicating findings to both technical and non-technical stakeholders.
- Worked with cross-functional engineering and client teams to translate performance findings into deployment and optimisation decisions for production systems.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.
- Applied image preprocessing and artefact-detection workflows with OpenCV and scikit-image to identify camera-feed contamination, reducing false positives by 13% in production vision pipelines.
- Improved production video inference reliability by adding structured logging, health checks, retries, and safe restart behaviour, reducing live pipeline crashes by 67% in 24/7 operation.
- Debugged training dynamics and convergence issues in production CV models, balancing model complexity, augmentation, and validation strategy to improve generalisation under lighting variation, occlusion, motion artefacts, and camera contamination.
- Analysed model performance and failure patterns from live video deployments, turning F1, false-positive and stability metrics into deployment guardrails and optimisation recommendations for engineering and operations teams.
- Presented model performance, failure modes, and deployment trade-offs to engineering teams, client stakeholders, and field operators, turning messy production signals into clear optimisation recommendations.
- Developed and maintained NVIDIA DeepStream 6.x video inference pipelines for 24/7 industrial perception deployments, supporting camera ingestion, model inference, and field diagnostics in harsh desert environments.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Led a team to design, train, and deploy vision pipelines for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries.
- Built low-latency asynchronous perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel hardware.
- Deployed quantised vision models using FP16/INT8 optimisation, pruning and layer fusion, reducing model size by 50% while preserving production accuracy targets.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using Python and OpenCV.
- Ran systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Led a 4-engineer team to design, train, and deploy CNN-based object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50% with Keras and Intel OpenVINO.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Collaborated with hardware, software, and QA teams to integrate CV models into deployed systems, and presented performance findings to leadership and clients to support rollout, optimisation, and reliability decisions.

- Engineered production-grade edge perception software for safety-critical environments, delivering deterministic real-time behaviour on Intel VPU devices.
- Mentored junior engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.
- Presented model performance, failure analysis and deployment trade-offs to engineering, QA, hardware and client teams; findings directly shaped rollout and optimisation decisions.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.
- Conducted optimisation experiments across quantisation-aware training, pruning, layer fusion, FP16/INT8 deployment and profiling, reducing model size by 50% while preserving accuracy under edge constraints.
- Implemented optical flow-based ego-motion estimation feeding motion representations into CNNs to classify vehicle state in production pipelines.
- Built asynchronous inference pipelines using OpenVINO and C++17 to maximise throughput on Intel VPU edge hardware under real-time latency and power constraints.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Designed data handling and retention practices aligned with GDPR principles, including data minimisation and controlled access, within production computer vision systems.
- Owned end-to-end production ML pipelines spanning data ingestion, labelling, training, optimisation, deployment, monitoring, and field validation for industrial safety systems, improving deployment reliability and release consistency.
- Provided technical leadership to a 4-member engineering team, setting experimentation standards, reviewing model performance, and guiding optimisation strategies for production deployments.
- Implemented an asynchronous inference pipeline in C++ with OpenVINO, overlapping pre-processing, inference, and post-processing to reduce end-to-end latency and improve pipeline utilisation on Intel GPU targets.
- Profiled and optimised C++ inference and pre and post-processing hotspots using Intel VTune Profiler for targeted performance fixes in our CNN inference pipeline.
- Integrated Boost C++ libraries into production pipelines, resolving dependency and build issues and maintaining reproducible deployments.
- Defined optimisation workflows spanning quantisation-aware training, pruning, layer fusion, FP16/INT8 deployment, and profiling, reducing model size by 50% while meeting accuracy, latency, and memory targets.
- Built automated data, labelling, and performance-tracking pipelines, repurposing older models for annotation and cutting manual labelling effort by 80% and also developed synthetic-data tooling to improve edge-case coverage and reduce manual labelling time by 60%.
- Worked closely with hardware, software, and QA teams to integrate perception modules into deployed systems and support rollout, optimisation, and reliability decisions.
- Optimised model architectures using quantisation-aware training, structured pruning, layer fusion, FP16/INT8 deployment, and targeted profiling, reducing model size by 50% while meeting accuracy, latency, and memory constraints.
- Owned end-to-end production ML pipelines from data ingestion and labelling through training, optimisation, deployment, monitoring, and field validation for industrial safety systems.
- Built a low-latency, high-reliability computer vision PoC for a zinc ore refining plant in South Africa, and developed a user-facing hazard detection application for an Oman-based client.
- Built and maintained a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising camera streams and running CV-based incident detection across both feeds under edge compute constraints.
- Implemented optical flow-based ego-motion estimation to feed motion representations into CNNs for vehicle-state classification in production pipelines.
- Explored semantic and instance segmentation models, including DeepLabv3+, Mask R-CNN, U-Net-style architectures, and SegFormer, to support annotation review, foreground/background separation, and failure analysis for industrial camera feeds.

- Built YOLOv5-based visual quality inspection for John Deere engine paintjobs on a conveyor line, detecting paint defects, contamination, and colour inconsistency as part of a manufacturing anomaly detection workflow.
- Used confusion-matrix failure clustering to diagnose recurring models failure, to prioritise dataset fixes, and improve model robustness across camera feeds.
- Integrated perception modules with hardware, software and QA teams, translating model performance, latency and failure analysis into rollout decisions for deployed safety systems.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% on large real-world datasets.
- Worked with hardware, software, QA, and client teams to translate model performance, latency, and failure analysis into rollout decisions for deployed safety systems.
- Tuned camera and ISP settings to keep models reliable on RGB+IR streams across 24/7 operation, cutting false positives on IR and reducing manual recalibration.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, VLP-16 and Intel RealSense D435i.
- Performed a JetPack platform upgrade on Jetson Xavier AGX to access newer CUDA and TensorRT, porting a ROS2/PCL SLAM stack across the upgrade and resolving the dependency and driver breakage it introduced.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **SLAM, Camera and Data Pipelines:** Camera ingestion, Cartographer, Ceres, Eigen, Filtering, GLIM, LiDAR SLAM, LIO-SAM, Multi-camera perception, Pose graph optimisation, Registration, RTAB-Map, Scan matching, Segmentation, Video and sensor data processing pipelines, Visual data streams
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), failure-mode analysis, Health checks, Intel VPUs, Microsoft Azure, MLflow, model-promotion gates, Post-deployment validation, Production monitoring, Structured logging
- **Programming & Tooling:** Bash, C++17, CMake, colcon, CUDA C++, Django, FastAPI, Git, Intel VTune, Linux, NumPy, OpenCV, Pandas, PG vector, PIL, PostgreSQL, Python, Qt, Scikit-image, Scikit-Learn
- **Edge Inference and Optimisation:** Bayesian Hyperparameter Optimisation, Convolutional Neural Networks, custom TensorRT plugins, dynamic batching, FP16/INT8 deployment, latency/throughput profiling, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, VRAM profiling, YOLO
- **Edge AI Hardware:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Embedded and Low-level:** 8051 Assembly, ADC/DAC interfacing, Arduino C, Embedded C, Interrupt-driven I/O, Timers and counters, UART serial
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image

- **Robotics and Middleware:** Closed-loop robot control, Gazebo Harmonic, Inverse kinematics, Nav2, rclcpp, rclpy, Real-time perception systems, robot_localization, ROS2 (Robot Operating System 2), rosbag2, sensor data playback, Sensor integration, tf2, timestamp synchronisation
- **Large Language Models:** Citation-grounded response generation, Error analysis, Experimentation and eval, Information extraction, Model fine-tuning, Retrieval-augmented generation (RAG), Search and Retrieval, Workflow automation
- **Computer Vision & Image Processing:** Anomaly detection, ByteTrack, Data augmentation, Defect detection, EfficientDet, Image processing, LiDAR extrinsics calibration, NvDCF, Object detection, Object tracking, RT-DETR, Segmentation, Sensor calibration, SSD MobileNet V2, Visual quality inspection, YOLO26, YOLOv8
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Prompt engineering
- **Computer Vision Models:** ByteTrack, DeepLabv3+, Depth Anything v2, EfficientViT, Faster R-CNN, Grounding DINO, Mask R-CNN, MobileSAM, SegFormer, U-Net, YOLOv10, YOLOv5
- **Datasets, Annotation & Evaluation:** Automated labelling, Cityscapes, COCO, Confusion-matrix-driven failure clustering, CVAT, Dataset versioning, DVC, F1-score, ImageNet, IoU, KITTI, Latency benchmarking, mAP, Model-size benchmarking, PASCAL VOC, Precision/Recall, Synthetic data generation
- **AI Evaluation:** citation checks, document-grounded QA, error analysis, evidence matching, hallucination detection, LLM-as-judge, RAG evaluation, retrieval metrics, structured outputs
- **Experimentation and Reliability:** ablation studies, benchmarking, failure-mode analysis, hypothesis testing, post-deployment validation, production monitoring, regression checks
- **Python and ML Engineering:** Apache Airflow, DVC, MLflow, NumPy, Optuna, Pandas, Python, PyTorch, scikit-learn
- **Production Systems:** Apache Kafka, CI/CD, Docker, Git, health checks, Linux, observability, structured logging
- **Applied AI:** anomaly detection, information extraction, object detection, ONNX, OpenCV, OpenVINO, YOLO
- **Edge AI & Video Pipelines:** camera behaviour debugging, CUDA, GStreamer, ISP tuning, JetPack, Jetson AGX Xavier, multi-stream video analytics, nvdsosd, NVIDIA DeepStream 6.x/9.0, nvinfer, nvstreammux, nvtracker, optical performance eval, RTSP video pipelines, TensorRT

AWARDS & SCHOLARSHIPS

Best Speaker award

ABC Toastmasters club

Jan 2020